

BazarHQ

Software Requirements Specification

Multi-Tenant SaaS E-Commerce Platform

Document Title	Software Requirements Specification — BazarHQ
Version	1.0
Status	Draft
Standard	IEEE 830
Date	April 2026

Prepared by the BazarHQ Development Team

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document defines the complete functional and non-functional requirements for BazarHQ – a multi-tenant Software as a Service (SaaS) e-commerce platform. The purpose of this document is to provide a clear, unambiguous, and comprehensive description of the system to be developed, serving as the primary reference for the development team, academic evaluators, and all project stakeholders.

This document is intended for the following audiences:

- Development Team – as the authoritative guide for what the system must do and how it must behave
- Academic Evaluators / Professor – as evidence of rigorous requirements engineering following the IEEE 830 standard
- Project Stakeholders – as a shared understanding of the system's scope, boundaries, and capabilities

1.2 Scope

Product Name: BazarHQ

BazarHQ is a cloud-based multi-tenant e-commerce platform that enables merchants in Bangladesh to create, customize, and operate their own online shops without any technical knowledge. Each merchant receives a unique subdomain upon shop creation (e.g. rahims-leather.bazarhq.com). Customers visit these shops to browse products and place orders. A Super Admin oversees and manages the entire platform.

What BazarHQ Does (Version 1 Scope):

- Allows merchants to register, create, and publish a fully functional online shop
- Provides storefront customization through pre-built themes
- Manages product catalogues with variants, images, and stock control
- Processes customer orders through Bangladeshi payment gateways
- Delivers order tracking for customers and order management for merchants
- Provides a super admin panel for platform-wide oversight

What BazarHQ Does Not Do (Deferred to Version 2):

- Custom domain support for individual merchant shops
- Multi-staff merchant accounts
- Direct courier and delivery partner integrations (Pathao, Steadfast)
- International payment gateways (Stripe, PayPal)
- Merchant subscription billing system
- Mobile application for merchants

1.3 Definitions, Acronyms & Abbreviations

Term	Definition
SRS	Software Requirements Specification

Term	Definition
SaaS	Software as a Service — cloud-hosted software accessed via a web browser
Multi-Tenancy	Architecture where multiple merchants share one platform but are fully data-isolated
Merchant	A business owner who registers on BazarHQ and operates an online shop
Customer	An end user who visits a merchant's storefront and places orders
Super Admin	A BazarHQ team member with full platform management access
Subdomain	A unique URL prefix assigned to each merchant shop (e.g. myshop.bazarhq.com)
RBAC	Role-Based Access Control — system access restrictions based on user role
JWT	JSON Web Token — a standard for secure, stateless authentication sessions
OAuth	Open Authorization — a standard for third-party authentication (e.g. Google login)
2FA	Two-Factor Authentication — a second verification step beyond password
TOTP	Time-Based One-Time Password — a 2FA method using an authenticator app
COD	Cash on Delivery — payment made in cash when the order is physically delivered
SSLCommerz	Bangladesh's leading payment gateway aggregator
bKash	Bangladesh's most widely used mobile financial service
Nagad	A Bangladesh postal service-backed mobile financial service
CDN	Content Delivery Network — distributed network for fast delivery of static assets
API	Application Programming Interface — communication interface between software systems
CSV	Comma-Separated Values — a spreadsheet file format for bulk data import/export
HTTPS	Hypertext Transfer Protocol Secure — encrypted web communication protocol
bcrypt	A password hashing algorithm used for secure password storage
FR	Functional Requirement
NFR	Non-Functional Requirement

1.4 References

- IEEE Std 830-1998 — IEEE Recommended Practice for Software Requirements Specifications
- Supabase Documentation — <https://supabase.com/docs>
- SSLCommerz Developer Documentation — <https://developer.sslcommerz.com>
- Cloudinary Documentation — <https://cloudinary.com/documentation>
- RFC 6238 — TOTP: Time-Based One-Time Password Algorithm
- Next.js Documentation — <https://nextjs.org/docs>

1.5 Overview

The remainder of this document is organized as follows:

- Section 2 provides an overall description of BazarHQ including product perspective, high-level functions, user characteristics, operating environment, and constraints.
- Section 3 defines all external systems and third-party dependencies that BazarHQ relies upon, their roles, and the impact of their unavailability.
- Section 4 defines all specific functional and non-functional requirements for the Merchant stakeholder, organized by feature module.
- Section 5 defines all specific functional and non-functional requirements for the Customer stakeholder, organized by feature module.
- Section 6 defines all specific functional and non-functional requirements for the Super Admin stakeholder, organized by feature module.
- Section 7 provides a full glossary of terms used throughout the document.

2. Overall Description

2.1 Product Perspective

BazarHQ is a new, self-contained SaaS platform. It is not a component of a larger existing system. It operates independently and interacts with the following external systems: Payment Gateways (SSLCommerz, bKash, Nagad) for processing customer payments; Email Gateway (Resend.com) for all transactional email communications; SMS Gateway (SSL Wireless or equivalent) for mobile notifications; and Cloud Storage (Cloudinary) for media file management and delivery.

The platform operates across three distinct URL spaces:

URL	Purpose
bazarhq.com	Platform landing page and merchant signup
{shop}.bazarhq.com	Customer-facing merchant storefront
{shop}.bazarhq.com/admin	Merchant admin panel (authenticated route)
bazarhq.com/superadmin	Super Admin panel (IP-restricted)

The system follows a multi-tenant architecture where all merchants share a single application and database infrastructure, with strict data isolation enforced at both the application and database level.

2.2 Product Functions

At a high level, BazarHQ provides the following major functions:

- Merchant Registration & Authentication — secure account creation and login
- Shop Creation & Onboarding — guided setup with subdomain assignment
- Storefront Customization — theme selection, branding, and layout configuration
- Product Management — full catalogue management with variants and bulk import
- Order Management — order receipt, status management, and invoice generation
- Payment Configuration — Bangladeshi gateway setup per merchant
- Analytics & Reporting — revenue, order, and visitor insights
- Customer Storefront — browsing, cart, checkout, and order tracking
- Super Admin Panel — platform-wide oversight, merchant management, and content control

2.3 User Classes & Characteristics

Primary Stakeholders:

Stakeholder	Technical Level	Usage Frequency	Primary Interface
Merchant	Low to Medium	Daily	Dashboard at {shop}.bazarhq.com/admin
Customer	Low	Occasional	Storefront at {shop}.bazarhq.com
Super Admin	High	Daily	Admin Panel at

Stakeholder	Technical Level	Usage Frequency	Primary Interface
			bazarhq.com/superadmin

Secondary Stakeholders (External Systems):

Stakeholder	Type	Role
Payment Gateway	External System	Financial transaction processing
Email Gateway	External System	Transactional email delivery
SMS Gateway	External System	Mobile notification delivery
Cloud Storage	External System	Media file storage and CDN delivery

2.4 Operating Environment

Frontend	Next.js (React) – server-side rendering for fast storefront load times
Backend	Node.js with Express – RESTful API
Database	PostgreSQL hosted on Supabase
Authentication	Supabase Auth with JWT session management
Hosting	Vercel (frontend), Railway.app (backend)
DNS & Subdomains	Cloudflare
Media Storage	Cloudinary
Target Devices	Mobile-first – minimum supported screen width 320px, full desktop support
Target Browsers	Chrome, Firefox, Safari, Samsung Internet (latest 2 versions)
Network	Designed for 3G mobile connections common in Bangladesh

2.5 Design & Implementation Constraints

ID	Constraint
CON-01	Platform built using free-tier cloud services – total monthly cost: ₳0
CON-02	Primary payment focus: SSLCommerz, bKash, Nagad, Cash on Delivery
CON-03	Mobile-first – desktop supported but mobile experience takes design priority
CON-04	Custom domain support for merchant shops is out of scope for Version 1
CON-05	Multi-staff merchant accounts are out of scope for Version 1
CON-06	Courier and delivery partner API integrations are out of scope for Version 1
CON-07	BazarHQ shall never store customer card numbers on its own servers
CON-08	All inter-service communication shall use HTTPS – no unencrypted HTTP in production
CON-09	One shop per merchant account limitation in Version 1

2.6 Assumptions & Dependencies

ID	Assumption / Dependency
ASM-01	Merchants have access to a smartphone and basic web browser literacy
ASM-02	Customers have access to a mobile phone capable of receiving SMS messages
ASM-03	SSLCommerz, bKash, and Nagad APIs are accessible from the deployment region
ASM-04	Supabase free tier (500MB database) is sufficient for Version 1 development
ASM-05	Resend.com free tier (100 emails/day) is sufficient during development
ASM-06	Cloudinary free tier (25GB storage, 25GB bandwidth) sufficient for early launch
ASM-07	Merchants have their own valid bKash and SSLCommerz credentials
ASM-08	Internet connectivity is available for all core operations — no offline support in V1
ASM-09	Bangladesh SMS delivery supports alphanumeric sender IDs via chosen provider

3. External Systems & Dependencies

S1

S1 – Payment Gateway

Provider

SSLCommerz, bKash, Nagad, Rocket

Role in the Platform:

Financial processing services that handle all monetary transactions. BazarHQ facilitates the connection without processing payments directly.

What BazarHQ Expects From This System:

- Stable API endpoint for initiating and verifying payment transactions
- Redirect mechanism to send customers to gateway's payment page and return them upon completion
- Success or failure response that BazarHQ uses to update order status
- Compliance with standard security protocols (HTTPS, token-based verification)

Impact if Unavailable:

- Card and mobile banking payments become unavailable at checkout
- Cash on Delivery remains functional as it has no external dependency
- System shall detect gateway unavailability and display a meaningful message to the customer

Constraints:

- BazarHQ shall never store customer card numbers or full mobile banking credentials
- Each merchant is responsible for maintaining their own valid gateway credentials
- BazarHQ cannot guarantee payment gateway uptime — it is outside the platform's control

S2

S2 – Email Gateway

Provider

Resend.com

Role in the Platform:

Delivers all transactional emails — merchant verification, password resets, order confirmations, status notifications, suspension notices, and broadcast announcements.

What BazarHQ Expects From This System:

- Reliable email delivery within 30 seconds of a send request under normal conditions
- Support for HTML formatted email templates

- Delivery status feedback so BazarHQ can detect failed deliveries
- Minimum 100 emails per day on the free tier during development

Impact if Unavailable:

- Merchants will not receive order notification emails
- Customers will not receive order confirmation emails
- Password reset flow will be broken for both merchants and customers
- System shall retry delivery up to 3 times before marking as failed; SMS gateway serves as fallback

Constraints:

- Email content must comply with anti-spam standards — all emails shall include unsubscribe option
- BazarHQ shall not send marketing emails through the transactional email gateway
- Daily email volume shall be monitored to avoid exceeding free tier limits

S3

S3 — SMS Gateway

Provider

SSL Wireless or equivalent Bangladeshi SMS provider

Role in the Platform:

Handles all SMS communications to customers and merchants. SMS is the primary real-time channel given Bangladesh's mobile-dominant user base.

What BazarHQ Expects From This System:

- Reliable SMS delivery to Bangladeshi phone numbers (01X format) within 30 seconds
- Support for alphanumeric sender ID so messages display as 'BazarHQ'
- Delivery receipt confirmation so BazarHQ can detect failed messages
- Reasonable cost per SMS compatible with free or low-cost tier during development

Impact if Unavailable:

- Customers will not receive order confirmation SMS after placing an order
- Customers will not receive order status update SMS
- Merchants will not receive new order SMS alerts
- System shall fall back to email automatically if SMS delivery fails

Constraints:

- SMS messages shall be maximum 160 characters to avoid multi-part SMS charges
- SMS shall only be sent for transactional events — no marketing messages in Version 1
- Customer phone numbers shall never be shared with the SMS gateway beyond message delivery

S4

S4 — Cloud Storage

Provider	Cloudinary (primary), Supabase Storage (fallback)
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Role in the Platform:

Handles all media files — shop logos, hero banners, product images, and profile pictures. BazarHQ does not store media files on its own servers.

What BazarHQ Expects From This System:

- Reliable file upload accepting JPG, PNG, and WEBP formats
- Automatic image compression and optimization upon upload
- CDN delivery of images for fast load times across Bangladesh
- Minimum 25GB storage and 25GB monthly bandwidth on free tier
- Stable and consistent URL structure for stored files

Impact if Unavailable:

- Merchants will be unable to upload new product images, logos, or banners
- Existing uploaded images will continue to display if CDN cache is available
- System shall display a clear error message to merchant if upload fails

Constraints:

- Individual file uploads shall not exceed 2MB — enforced client-side before upload attempt
- BazarHQ shall not store sensitive documents or PII in cloud storage — only media assets
- Stored file URLs are semi-public — system shall not rely on URL obscurity for security

External Systems Summary:

ID	System	Provider	Critical Dependency	Fallback
S1	Payment Gateway	SSLCommerz, bKash, Nagad	Checkout & payment processing	Cash on Delivery
S2	Email Gateway	Resend.com	All transactional emails	SMS Gateway
S3	SMS Gateway	SSL Wireless / Twilio	Order alerts & notifications	Email Gateway
S4	Cloud Storage	Cloudinary / Supabase	Media file storage & delivery	None in V1

4. Specific Requirements — Merchant

M1

M1 — Registration & Authentication

Purpose:
This section defines how a merchant creates an account on BazarHQ and securely accesses their dashboard. This is the entry point of the entire merchant journey.

- Actors Involved:**
- Primary: Merchant
 - Secondary: BazarHQ Authentication System (Supabase Auth)

Functional Requirements

- Registration**
- The system shall allow a merchant to register using their email address and password
 - The system shall allow a merchant to register using their Google account (OAuth)
 - The system shall send a verification email upon successful registration
 - The system shall not allow registration with an email address that already exists in the system
 - The system shall validate password strength — minimum 8 characters with at least one number

- Login**
- The system shall allow a merchant to log in using email and password
 - The system shall allow a merchant to log in using Google OAuth
 - The system shall redirect the merchant to their dashboard upon successful login
 - The system shall display a clear error message if credentials are incorrect
 - The system shall lock the account temporarily after 5 consecutive failed login attempts

- Session Management**
- The system shall maintain a merchant session using JWT tokens
 - The system shall expire the session after 7 days of inactivity
 - The system shall allow a merchant to log out from all active devices simultaneously

Non-Functional Requirements

- Passwords shall be hashed using bcrypt before storage — never stored as plain text
- Login response time shall not exceed 2 seconds under normal load
- All authentication traffic shall be encrypted via HTTPS
- Failed login attempts shall be logged with timestamp and IP address

Constraints & Edge Cases

- If a merchant registers via Google OAuth, they cannot set a password unless they use the Forgot Password flow
- A merchant with a suspended account shall see a clear suspension message upon login — not a generic error
- Email verification must be completed before the merchant can access the dashboard or create a shop

M2**M2 — Shop Creation & Onboarding****Purpose:**

This section defines the process by which a verified merchant creates their shop on BazarHQ, selects their unique identity, and completes the initial onboarding steps before their storefront goes live.

Actors Involved:

- Primary: Merchant
- Secondary: BazarHQ Shop Management System

Functional Requirements**Shop Creation**

- The system shall prompt the merchant to create a shop immediately after email verification
- The system shall allow the merchant to enter a shop name
- The system shall automatically generate a subdomain based on the shop name (e.g. rahims-leather.bazarhq.com)
- The system shall check subdomain availability in real time as the merchant types
- The system shall not allow two shops to share the same subdomain
- The system shall allow the merchant to manually edit the suggested subdomain before confirming
- The system shall only accept subdomains containing lowercase letters, numbers, and hyphens
- The system shall save the shop as a draft until the merchant explicitly publishes it

Onboarding Flow

- The system shall guide the merchant through a step-by-step onboarding checklist after shop creation
- The onboarding checklist shall include: customize shop, add first product, configure payment, publish shop
- The system shall display onboarding progress as a percentage on the dashboard
- The system shall allow the merchant to skip onboarding steps and return to them later
- The system shall mark the onboarding as complete once all steps are finished

Shop Profile

- The system shall allow the merchant to enter a shop description during onboarding
- The system shall allow the merchant to set a business contact email and phone number
- The system shall allow the merchant to set the shop's primary business category (e.g. Fashion, Electronics, Food, Handicrafts)

Non-Functional Requirements

- Subdomain availability check shall respond within 1 second
- Shop creation shall complete and redirect to the dashboard within 3 seconds

- Subdomain uniqueness shall be enforced at the database level with a unique constraint
- Shop data shall be isolated — no other merchant shall be able to read or modify another merchant's data

Constraints & Edge Cases

- A single merchant account shall be limited to one shop in Version 1
- If the merchant closes the browser during onboarding, their progress shall be saved and resumed on next login
- Reserved words such as 'admin', 'superadmin', 'api', 'www', and 'support' shall not be permitted as subdomains
- Deleting a shop shall require explicit confirmation and shall permanently remove all associated products, orders, and data

M3**M3 — Store Customization****Purpose:**

This section defines how a merchant personalizes the visual appearance and content of their storefront. The goal is to allow meaningful branding without requiring any technical or design knowledge.

Actors Involved:

- Primary: Merchant
- Secondary: BazarHQ Theme Engine, Cloudinary (image storage)

Functional Requirements**Theme Selection**

- The system shall provide a minimum of 4 pre-built storefront themes for the merchant to choose from
- The system shall display a live preview of each theme before the merchant applies it
- The system shall allow the merchant to switch themes at any time without losing their content
- The system shall apply the selected theme to the customer-facing storefront immediately upon saving

Branding

- The system shall allow the merchant to upload a shop logo (PNG or JPG, maximum 2MB)
- The system shall allow the merchant to set a primary brand color using a color picker
- The system shall allow the merchant to select a font style from a predefined list of minimum 3 options
- The system shall allow the merchant to upload a hero banner image for the storefront homepage
- The system shall allow the merchant to write a homepage tagline of maximum 100 characters

Layout & Sections

- The system shall allow the merchant to toggle individual homepage sections on or off (hero banner, featured products, about section, announcement bar)
- The system shall allow the merchant to choose between grid view and list view for product display
- The system shall allow the merchant to select which products appear in the featured products section
- The system shall allow the merchant to write and edit an About Us section for their shop

Content & Policies

- The system shall allow the merchant to write a return policy page
- The system shall allow the merchant to write a shipping policy page
- The system shall allow the merchant to set an announcement bar message (e.g. Free delivery over ₪1000)
- The system shall allow the merchant to add social media links (Facebook, Instagram, WhatsApp)
- The system shall allow the merchant to set footer contact information

Non-Functional Requirements

- Theme preview shall load within 2 seconds
- Logo and banner images shall be automatically compressed and served via CDN upon upload
- All customization changes shall reflect on the live storefront within 5 seconds of saving
- The storefront shall remain accessible to customers during customization changes – no downtime shall occur

Constraints & Edge Cases

- Uploaded images must not exceed 2MB – the system shall display a clear error if this limit is exceeded
- If no logo is uploaded, the system shall display the shop name as text in the header by default
- If no banner image is uploaded, the system shall use the selected theme's default banner
- The color picker shall default to the theme's recommended color palette to guide non-designer merchants

M4**M4 — Product Management****Purpose:**

This section defines how a merchant adds, organizes, edits, and manages the products available in their storefront. Products are the core inventory of the shop and directly determine what customers can browse and purchase.

Actors Involved:

- Primary: Merchant
- Secondary: Cloudinary (image storage), BazarHQ Inventory System

Functional Requirements**Adding Products**

- The system shall allow the merchant to add a new product with mandatory fields: name, description, price, category, and stock quantity
- The system shall allow the merchant to upload a minimum of 1 and maximum of 6 images per product
- The system shall allow the merchant to designate one image as the primary thumbnail image
- The system shall allow the merchant to assign a product to a category (existing or newly created)
- The system shall allow the merchant to set a compare-at price (original price) for displaying discounts
- The system shall allow the merchant to add product tags for easier search and filtering

Product Variants

- The system shall allow the merchant to define variants for a product (e.g. Size: S, M, L, XL or Color: Red, Blue)
- The system shall allow the merchant to set individual stock quantities per variant
- The system shall allow the merchant to set individual prices per variant where they differ from the base price
- The system shall allow the merchant to define multiple variant types on a single product (e.g. Size AND Color)

Managing Products

- The system shall allow the merchant to edit any product detail at any time
- The system shall allow the merchant to set a product status as Active, Draft, or Archived
- The system shall allow the merchant to delete a product permanently
- The system shall display a warning before permanent deletion if the product has associated orders
- The system shall allow the merchant to duplicate an existing product to speed up adding similar items
- The system shall display a low stock warning when any product or variant quantity falls below a merchant-defined threshold

Bulk Operations

- The system shall allow the merchant to import products in bulk via a CSV file
- The system shall validate the CSV file and display errors for any rows with missing or invalid data before processing
- The system shall allow the merchant to export their full product list as a CSV file
- The system shall support a maximum of 500 products per CSV import

Categories

- The system shall allow the merchant to create, rename, and delete product categories
- The system shall allow the merchant to assign multiple products to a category simultaneously
- The system shall display categories as filter options on the customer-facing storefront

Non-Functional Requirements

- Product images shall be compressed and optimized automatically upon upload
- Product pages on the storefront shall load within 3 seconds on a standard mobile connection
- CSV import of 500 products shall complete within 30 seconds
- Product stock count shall update in real time when an order is placed to prevent overselling
- The product management interface shall support a catalogue of up to 1000 products per shop without performance degradation

Constraints & Edge Cases

- A product must have at least a name and a price before it can be saved
- If a product has active orders associated with it, it cannot be deleted — only archived
- If a variant runs out of stock, it shall be automatically marked unavailable while the product remains visible
- CSV imports shall not overwrite existing products by default — only add new ones unless merchant selects update mode

M5**M5 — Order Management****Purpose:**

This section defines how a merchant receives, views, processes, and fulfills customer orders. Order management is the operational heartbeat of the merchant's daily workflow on BazarHQ.

Actors Involved:

- Primary: Merchant
- Secondary: Customer, BazarHQ Notification System, SMS/Email Gateway

Functional Requirements**Order Notifications**

- The system shall notify the merchant via email when a new order is placed
- The system shall notify the merchant via SMS when a new order is placed
- The system shall display a real-time notification badge on the dashboard when a new order arrives
- The system shall allow the merchant to configure their preferred notification method (email, SMS, or both)

Order List View

- The system shall display all orders in a list with: order ID, customer name, date, total, payment method, and status
- The system shall allow the merchant to filter orders by status (Pending, Confirmed, Shipped, Delivered, Cancelled)
- The system shall allow the merchant to search orders by order ID or customer name
- The system shall allow the merchant to sort orders by date or total amount
- The system shall display orders with the most recent at the top by default

Order Detail View

- The system shall allow the merchant to click any order to view its full details
- The order detail page shall display: customer name, phone, address, items with quantities and prices, payment method, payment status, and full order timeline
- The system shall display the complete order timeline showing when each status change occurred

Order Status Management

- The system shall allow the merchant to update order status: Pending -> Confirmed -> Shipped -> Delivered
- The system shall allow the merchant to cancel an order and specify a cancellation reason
- The system shall automatically notify the customer via SMS or email when their order status is updated
- The system shall not allow the merchant to revert an order to a previous status once updated

Invoice

- The system shall allow the merchant to generate and download a PDF invoice for any order
- The invoice shall include: shop name and logo, order ID, customer details, itemized product list, subtotal, delivery charge, and total

Non-Functional Requirements

- The order list shall load within 2 seconds even with 1000+ orders in the database
- SMS and email notifications shall be delivered within 30 seconds of order placement
- Order status updates shall be reflected on the customer's tracking page within 5 seconds
- The system shall guarantee no order is lost during high traffic — all orders shall be queued and processed reliably

Constraints & Edge Cases

- An order marked as Delivered cannot be cancelled — merchant must resolve directly with the customer
- If SMS delivery fails, the system shall fall back to email notification automatically
- A Pending order with no merchant action for 48 hours shall trigger a reminder notification to the merchant
- Cash on Delivery orders shall have payment status 'Pending' until the merchant manually marks it as 'Collected'

M6**M6 — Payment Configuration****Purpose:**

This section defines how a merchant sets up and manages the payment methods available to their customers at checkout. Merchants configure their own gateway credentials, and BazarHQ facilitates the connection without storing sensitive financial data.

Actors Involved:

- Primary: Merchant
- Secondary: SSLCommerz, bKash, Nagad, BazarHQ Payment Module

Functional Requirements**Available Payment Methods**

- The system shall provide the following payment methods for merchants to enable: bKash, Nagad, Rocket, SSLCommerz (cards and net banking), and Cash on Delivery
- The system shall allow the merchant to enable or disable any payment method at any time
- The system shall require at least one payment method to be active before the shop can be published
- The system shall display only the merchant's enabled payment methods to customers at checkout

Gateway Configuration

- The system shall allow the merchant to enter their bKash merchant number for bKash payments
- The system shall allow the merchant to enter their Nagad merchant number for Nagad payments
- The system shall allow the merchant to enter their SSLCommerz Store ID and Store Password for card payments
- The system shall mask sensitive credentials on the UI after saving — displaying only the last 4 characters
- The system shall validate that entered credentials are in the correct format before saving

Cash on Delivery

- The system shall allow the merchant to enable Cash on Delivery as a payment option
- The system shall allow the merchant to restrict COD to specific delivery zones if needed
- The system shall allow the merchant to set a maximum order value eligible for COD

Payment Records

- The system shall display a payment summary per order showing: payment method, amount, and payment status (Paid, Pending, Failed)
- The system shall allow the merchant to manually mark a COD order as 'Payment Collected' upon delivery

Non-Functional Requirements

- Payment credentials shall be stored encrypted in the database — never in plain text
- BazarHQ shall never store customer card details — all card processing handled entirely by SSLCommerz
- Payment gateway connections shall timeout gracefully within 10 seconds with a meaningful error displayed
- All payment-related API calls shall be made exclusively over HTTPS

Constraints & Edge Cases

- If a merchant disables a payment method, previous orders using that method remain unaffected
- A merchant cannot process refunds through BazarHQ in Version 1 — refunds handled directly between merchant and customer
- If SSLCommerz credentials are invalid, system shall disable card payments and notify the merchant — not expose raw gateway errors
- Merchants are solely responsible for the accuracy of their payment credentials

M7

M7 — Analytics & Reporting

Purpose:

This section defines the data insights available to the merchant through their dashboard. Analytics help merchants understand their shop performance, track revenue trends, and make informed business decisions.

Actors Involved:

- Primary: Merchant
- Secondary: BazarHQ Analytics Engine

Functional Requirements

Revenue Analytics

- The system shall display total revenue for the current day, week, and month
- The system shall display a revenue trend chart showing daily revenue for the past 30 days
- The system shall display total revenue broken down by payment method (bKash, COD, card, etc.)
- The system shall highlight the highest revenue day within the selected period

Order Analytics

- The system shall display total order count for the current day, week, and month
- The system shall display a breakdown of orders by status (Pending, Confirmed, Shipped, Delivered, Cancelled)
- The system shall display the average order value for the selected period
- The system shall display the number of abandoned carts — sessions where a cart was created but no order was placed

Product Analytics

- The system shall display a list of top-selling products ranked by units sold
- The system shall display a list of top-selling products ranked by revenue generated
- The system shall display products with low or zero remaining stock in a dedicated alert section

Visitor Analytics

- The system shall display the total number of visitors to the shop per day for the past 30 days
- The system shall display the conversion rate — percentage of visitors who placed an order
- The system shall display which pages on the storefront receive the most traffic

Reporting

- The system shall allow the merchant to export an order report as a CSV file for a selected date range
- The system shall allow the merchant to export a revenue summary as a CSV file for a selected date range

Non-Functional Requirements

- The analytics dashboard shall load within 3 seconds
- Analytics data shall be updated within 5 minutes of any new order or visitor event
- Visitor tracking shall use a lightweight, privacy-respecting solution that does not negatively impact storefront load time
- Analytics data shall be retained for a minimum of 12 months per merchant

Constraints & Edge Cases

- Analytics data is read-only — the merchant cannot edit or delete any recorded data
- If a shop has zero orders, the dashboard shall display empty state illustrations with guidance rather than blank graphs
- Visitor count tracks unique sessions — a single visitor refreshing multiple times counts as one visit
- Cancelled orders shall be excluded from revenue calculations but included in order count with a 'Cancelled' label

M8**M8 — Account Settings****Purpose:**

This section defines the personal account management options available to a merchant. This covers security settings, profile updates, notification preferences, and account lifecycle actions such as password change and account deletion.

Actors Involved:

- Primary: Merchant
- Secondary: BazarHQ Authentication System, Email/SMS Gateway

Functional Requirements**Profile Management**

- The system shall allow the merchant to update their full name
- The system shall allow the merchant to update their account email address after verifying the new email
- The system shall allow the merchant to update their contact phone number
- The system shall allow the merchant to upload a profile picture

Password & Security

- The system shall allow the merchant to change their password by entering their current and new password
- The system shall allow the merchant to reset their password via a time-limited link sent to their registered email
- The system shall allow the merchant to enable Two-Factor Authentication (2FA) using an authenticator app
- The system shall allow the merchant to view a list of active login sessions with device type and location info
- The system shall allow the merchant to terminate any specific active session remotely
- The system shall allow the merchant to terminate all sessions except the currently active one

Notification Preferences

- The system shall allow the merchant to choose their preferred notification channel: email, SMS, or both
- The system shall allow the merchant to toggle specific notification types on or off (new order, low stock alert, order status update)

Account Deletion

- The system shall allow the merchant to permanently delete their account
- The system shall require the merchant to type a confirmation phrase before account deletion proceeds

- The system shall inform the merchant that deletion is permanent and all shop data, products, and order history will be removed
- The system shall unpublish the merchant's shop immediately upon account deletion initiation
- The system shall complete account deletion within 30 days in compliance with data privacy best practices

Non-Functional Requirements

- Password reset links shall expire after 1 hour of generation
- 2FA setup shall follow the TOTP standard (RFC 6238) compatible with Google Authenticator and Authy
- All account setting changes shall be confirmed with a success notification within 2 seconds
- Account deletion shall be irreversible — no data recovery possible after the 30-day grace period

Constraints & Edge Cases

- A merchant who signed up via Google OAuth shall not have a password — 'Change Password' option shall be hidden with an explanatory note
- If a merchant has active pending or confirmed orders at the time of deletion request, the system shall warn them and recommend fulfilling orders first
- Email change shall require verification of the new email before taking effect — old email remains active until new one is verified
- If 2FA is enabled and the merchant loses access to their authenticator app, recovery is only possible via backup code provided during 2FA setup

5. Specific Requirements — Customer

C1

C1 — Shop Discovery & Browsing

Purpose:

This section defines how a customer finds and explores a merchant's storefront on BazarHQ. Since customers do not need an account to browse, this is the first touchpoint between the customer and the merchant's shop.

Actors Involved:

- Primary: Customer
- Secondary: BazarHQ Storefront Engine, Merchant Shop Data

Functional Requirements

Accessing the Shop

- The system shall allow any customer to visit a merchant's storefront via its unique subdomain without requiring login or account creation
- The system shall display the merchant's shop only if the shop is in a published state
- The system shall display a friendly 'Shop Not Found' page if the subdomain does not exist
- The system shall display a 'Shop Currently Unavailable' page if the merchant has unpublished their shop
- The system shall load the storefront in the merchant's selected theme and branding configuration

Homepage

- The system shall display the shop's hero banner, tagline, and announcement bar on the homepage if configured
- The system shall display a featured products section on the homepage if the merchant has enabled it
- The system shall display the shop's active product categories for easy navigation
- The system shall display the shop's contact information and social media links in the footer

Product Browsing

- The system shall display all active products in the merchant's catalogue in the layout configured by the merchant
- The system shall allow the customer to browse products by category
- The system shall allow the customer to filter products by price range
- The system shall allow the customer to sort products by newest, price low to high, and price high to low
- The system shall display each product card with thumbnail image, name, price, and compare-at price if applicable
- The system shall clearly mark out-of-stock products and prevent them from being added to the cart

Product Detail Page

- The system shall allow the customer to click any product to view its full detail page
- The product detail page shall display all product images in a scrollable gallery
- The product detail page shall display the product name, full description, price, available variants, and stock status
- The system shall allow the customer to select a variant before adding to cart
- The system shall update the displayed price and stock status dynamically when a variant is selected
- The system shall display a 'Only X left in stock' warning when stock quantity falls below 5 units

Search

- The system shall provide a search bar on the storefront for customers to search products by name or keyword
- The system shall display search results in real time as the customer types
- The system shall display a 'No products found' message with a suggestion to browse categories if search yields no results

Non-Functional Requirements

- The storefront homepage shall load within 3 seconds on a standard mobile connection
- Product search results shall appear within 1 second of the customer typing
- The storefront shall be fully responsive and usable on all screen sizes from 320px width upward
- Product images shall be served via CDN for fast load times regardless of location within Bangladesh
- The storefront shall function correctly on low-end Android devices commonly used in Bangladesh

Constraints & Edge Cases

- Draft and archived products shall never be visible to the customer regardless of direct URL access attempts
- If all products in a category are out of stock, the category remains visible but all products shall be clearly marked unavailable
- If the merchant has not uploaded any products, the storefront shall display a 'Coming Soon' message rather than a blank page
- If the shop is unpublished, the storefront URL shall not reveal any shop data or confirm the shop's existence

C2**C2 — Cart Management****Purpose:**

This section defines how a customer builds and manages their shopping cart before proceeding to checkout. The cart must be reliable, intuitive, and persistent enough to handle common real-world behaviors.

Actors Involved:

- Primary: Customer
- Secondary: BazarHQ Cart System, Merchant Inventory

Functional Requirements**Adding to Cart**

- The system shall allow the customer to add a product to the cart from both the listing page and the product detail page
- The system shall require the customer to select all available variants before the 'Add to Cart' button becomes active
- The system shall display a confirmation indicator when a product is successfully added to the cart
- The system shall not allow the customer to add an out-of-stock product or variant to the cart
- The system shall not allow the customer to add a quantity greater than the available stock

Viewing the Cart

- The system shall display a cart icon with an item count badge visible on all storefront pages
- The system shall allow the customer to open a cart summary panel or navigate to a dedicated cart page
- The cart shall display each item with its image, name, selected variant, unit price, quantity, and line total
- The system shall display the cart subtotal, estimated delivery charge, and order total at the bottom

Modifying the Cart

- The system shall allow the customer to increase or decrease the quantity of any item in the cart
- The system shall prevent the customer from setting a quantity higher than available stock
- The system shall allow the customer to remove any individual item from the cart
- The system shall allow the customer to clear the entire cart at once
- The system shall automatically update the subtotal and order total whenever the cart is modified

Cart Persistence

- The system shall persist the customer's cart using browser local storage so it is not lost upon page refresh
- The system shall retain cart contents for a minimum of 7 days without requiring a customer account
- The system shall clear the cart automatically and completely upon successful order placement

Non-Functional Requirements

- Cart updates shall reflect instantly without requiring a full page reload
- Cart total calculations shall be accurate to 2 decimal places at all times
- Cart state shall be consistent — same browser and device within 7 days shall have the cart intact
- The cart shall handle a maximum of 50 line items without performance degradation

Constraints & Edge Cases

- If a product's price changes after being added to cart, the cart shall update and display a notice to the customer before checkout
- If a product becomes out of stock after being added to cart, the system shall alert the customer at checkout and prevent proceeding until resolved
- Cart data is stored client-side — clearing browser data or switching devices means the cart is not recoverable without an account
- A guest customer's cart shall not be visible to the merchant until an order is formally placed

C3**C3 — Checkout & Payment****Purpose:**

This section defines the complete checkout process a customer goes through to place an order. The checkout must be fast, minimal, and fully optimized for mobile users making payments via Bangladeshi payment methods.

Actors Involved:

- Primary: Customer
- Secondary: Merchant Payment Configuration, SSLCommerz, bKash, Nagad, BazarHQ Order System

Functional Requirements**Checkout Initiation**

- The system shall allow the customer to proceed to checkout from the cart page
- The system shall present a checkout flow completable in a maximum of 3 steps
- The system shall not require the customer to create an account to complete checkout
- The system shall display an order summary throughout the entire checkout process

Step 1 — Delivery Information

- The system shall require the customer to enter: full name, phone number, full delivery address, and district
- The system shall allow the customer to optionally enter apartment number, area specification, and a delivery note
- The system shall validate that the phone number is in a valid Bangladeshi format (01XXXXXXXXXX)
- The system shall validate that no mandatory field is empty before allowing progression to the next step

Step 2 — Payment Method Selection

- The system shall display only the payment methods that the merchant has enabled for their shop
- The system shall allow the customer to select one payment method from the available options
- The system shall display clear instructions for the selected payment method
- The system shall allow the customer to enter their bKash or Nagad transaction ID after completing payment
- The system shall redirect the customer to the SSLCommerz payment gateway page if card payment is selected
- The system shall handle the SSLCommerz payment response and redirect the customer back to BazarHQ

Step 3 — Review & Place Order

- The system shall display a complete order summary for review: items, delivery address, payment method, and total

- The system shall display links to the merchant's return and shipping policy on the review page
- The system shall allow the customer to go back and edit any previous step before placing the order
- The system shall place the order only when the customer explicitly clicks the 'Place Order' button
- The system shall display a clear success page with the order ID immediately after successful placement

Post-Order

- The system shall send an order confirmation SMS to the customer's phone number within 30 seconds
- The system shall send an order confirmation email if the customer provided an email address
- The confirmation message shall include order ID, items ordered, total amount, and a tracking link

Non-Functional Requirements

- The full checkout flow shall be completable within 3 minutes by a first-time mobile user
- Checkout pages shall load within 2 seconds on a standard mobile connection
- Payment data shall never pass through or be stored on BazarHQ servers
- The system shall handle simultaneous checkouts from multiple customers without order conflicts or data mixing
- SSLCommerz redirect and return handling shall complete within 10 seconds under normal network conditions

Constraints & Edge Cases

- If SSLCommerz is unavailable, the system shall display a clear error and suggest alternatives — never expose raw gateway errors
- If a customer's session expires mid-checkout, their cart shall be preserved and they shall return to the beginning
- If stock runs out for an item between cart and checkout, the system shall notify the customer and prevent order placement
- For bKash and Nagad, the order shall be placed in 'Payment Pending Verification' status until the merchant confirms
- A customer cannot modify or cancel an order after successful placement — they must contact the merchant directly

C4**C4 — Order Tracking****Purpose:**

This section defines how a customer monitors the progress of their placed order. The experience must be simple, clear, and accessible without requiring an account.

Actors Involved:

- Primary: Customer
- Secondary: Merchant (order status updates), BazarHQ Order System, SMS Gateway

Functional Requirements**Accessing Order Tracking**

- The system shall provide a dedicated order tracking page at the merchant's storefront (e.g. rahims-leather.bazarhq.com/track)
- The system shall allow the customer to look up their order by entering their Order ID and registered phone number
- The system shall include the tracking page link in the order confirmation SMS and email
- The system shall not require the customer to have an account to access the order tracking page

Order Status Display

- The system shall display the current order status clearly at the top of the tracking page
- The system shall display a visual timeline: Order Placed, Payment Confirmed, Order Confirmed, Shipped, Delivered
- The system shall highlight completed stages and clearly indicate the current active stage
- The system shall display the date and time when each stage was reached
- The system shall display the estimated delivery timeframe if the merchant has configured one

Status Update Notifications

- The system shall send an SMS to the customer each time the merchant updates the order status
- The system shall include the new status and a tracking link in each SMS notification
- The system shall allow the customer to opt out of further SMS notifications from the tracking page

Order Summary on Tracking Page

- The system shall display the full order summary: ordered items, quantities, total amount, payment method, and delivery address
- The system shall display the merchant's shop name and contact information so the customer can reach out if needed

Non-Functional Requirements

- The order tracking page shall load within 2 seconds
- Order status shall reflect the merchant's latest update within 5 seconds of the merchant saving the change
- The tracking page shall be fully functional on basic mobile browsers without any app installation
- SMS notifications shall be delivered within 30 seconds of a status change being saved by the merchant

Constraints & Edge Cases

- If a customer enters an incorrect Order ID or phone number, the system shall display 'Order not found' — not reveal whether the Order ID exists
- If the merchant has not updated order status for 48+ hours, the system shall display the last known status without implying delay
- A delivered order's tracking page shall remain accessible for 90 days after delivery before being archived
- The tracking page shall display a clear message if the order was cancelled, including the merchant's cancellation reason

C5

C5 — Customer Account (Optional)

Purpose:

This section defines the optional account system available to customers. A customer account is not required to shop on BazarHQ — it is an enhancement providing convenience features like order history and saved addresses.

Actors Involved:

- Primary: Customer
- Secondary: BazarHQ Authentication System

Functional Requirements

Account Creation

- The system shall allow a customer to create an optional account using their email address and password
- The system shall allow a customer to register using their Google account (OAuth)
- The system shall send a verification email upon registration
- The system shall allow a guest customer to create an account after placing an order by simply setting a password

Login & Session

- The system shall allow the customer to log in using email and password
- The system shall allow the customer to log in using Google OAuth
- The system shall maintain the customer session for 30 days using a secure token
- The system shall allow the customer to log out at any time

Order History

- The system shall display a list of all past orders placed by the logged-in customer across any BazarHQ shop
- Each order shall display: order ID, shop name, date, items, total amount, and current status
- The system shall allow the customer to click any past order to view its full details and tracking status

Saved Information

- The system shall allow the customer to save up to 3 delivery addresses for faster checkout
- The system shall allow the customer to set a default delivery address that pre-fills at checkout
- The system shall allow the customer to update or delete any saved address

Account Settings

- The system shall allow the customer to update their name and phone number

- The system shall allow the customer to change their password by entering their current and new password
- The system shall allow the customer to reset their password via a time-limited email link
- The system shall allow the customer to permanently delete their account
- The system shall inform the customer that account deletion removes their saved data but does not affect merchant order records

Non-Functional Requirements

- Customer account data shall be completely separate from merchant account data at the database level
- Customer passwords shall be hashed using bcrypt before storage
- Customer session tokens shall expire after 30 days of inactivity
- Account creation shall complete and redirect within 3 seconds

Constraints & Edge Cases

- A customer account is platform-scoped – one account covers all BazarHQ shops without separate accounts per shop
- If a guest customer places an order with an email already registered, the system shall associate the order with that account and notify the customer
- A customer who deletes their account loses access to order history on the platform – however merchants retain their order data
- If a customer registered via Google OAuth, the 'Change Password' option shall be hidden from their account settings

6. Specific Requirements — Super Admin

A1

A1 — Admin Authentication

Purpose:

This section defines how a BazarHQ Super Admin securely accesses the platform's administration panel. Admin authentication demands a significantly higher level of security given the admin's ability to affect the entire platform.

Actors Involved:

- Primary: Super Admin (BazarHQ Team Member)
- Secondary: BazarHQ Authentication System, 2FA Authenticator App

Functional Requirements

Login

- The system shall provide a separate admin login page accessible at bazarhq.com/superadmin
- The system shall allow the super admin to log in using an email address and password
- The system shall not display any public-facing sign-up option — admin accounts shall only be created internally
- The system shall require the super admin to complete a Two-Factor Authentication (2FA) step after entering valid credentials
- The system shall use a time-based one-time password (TOTP) challenge via an authenticator app as the 2FA method
- The system shall deny access if the 2FA code is incorrect or has expired
- The system shall redirect the super admin to the admin dashboard upon successful authentication

Failed Login Handling

- The system shall lock the admin account for 30 minutes after 3 consecutive failed login attempts
- The system shall send an email alert to the registered admin email address upon account lockout
- The system shall log every failed login attempt with timestamp, IP address, and user agent

Session Management

- The system shall maintain the admin session for a maximum of 8 hours regardless of activity
- The system shall terminate the admin session automatically after 30 minutes of inactivity
- The system shall allow the super admin to manually log out at any time
- The system shall invalidate all active admin sessions immediately upon password change

Access Control

- The system shall restrict the admin panel to whitelisted IP addresses in production
- The system shall display a blocked access page with no technical details for non-whitelisted IPs
- The system shall support multiple super admin accounts with individual credentials
- The system shall assign roles to admin accounts: Viewer (read-only) and Full Access (complete management control)

Non-Functional Requirements

- All admin panel traffic shall be encrypted via HTTPS at all times
- The admin login page shall not be indexed by search engines – robots.txt and meta tags shall prevent crawling
- Admin session tokens shall be stored using httpOnly cookies to prevent JavaScript access
- All authentication events shall be written to an immutable audit log
- The 2FA TOTP implementation shall comply with RFC 6238 standards

Constraints & Edge Cases

- If a super admin loses access to their 2FA app, account recovery is only possible through an internal process managed by the development team
- Admin accounts cannot be created or deleted through the admin panel UI in Version 1 – only at database level by the development team
- If IP whitelisting is disabled for any reason, an alert shall be logged and the development team notified immediately
- A super admin shall not be able to view or modify their own audit log entries – another team member must review them

A2**A2 — Merchant Management****Purpose:**

This section defines how the super admin views, monitors, and manages all merchant accounts registered on the BazarHQ platform. This is the most operationally sensitive section of the admin panel as actions taken here directly affect active merchants.

Actors Involved:

- Primary: Super Admin
- Secondary: Merchant Accounts, BazarHQ Notification System

Functional Requirements**Merchant List View**

- The system shall display a paginated list of all registered merchants on the platform
- Each entry shall display: merchant name, email, shop name, subdomain, registration date, shop status, and account status
- The system shall allow the super admin to search merchants by name, email, or shop name
- The system shall allow the super admin to filter merchants by account status, shop status, and registration date range
- The system shall allow the super admin to sort merchants by registration date, shop name, or account status
- The system shall display the total number of registered merchants at the top of the list

Merchant Detail View

- The system shall allow the super admin to click any merchant to view their full profile
- The detail page shall display: personal details, shop info, total products, total orders, total revenue, payment methods configured, and activity log
- The system shall display the merchant's shop status and a direct link to their live storefront

Merchant Account Actions

- The system shall allow the super admin to suspend a merchant account
- The system shall require the super admin to enter a reason before suspending a merchant account
- The system shall immediately log the merchant out of all active sessions upon suspension
- The system shall automatically unpublish the merchant's shop upon account suspension
- The system shall send an email notification to the merchant informing them of the suspension and the stated reason
- The system shall allow the super admin to reinstate a previously suspended merchant account
- The system shall send an email notification to the merchant upon reinstatement
- The system shall allow the super admin to permanently delete a merchant account after explicit confirmation

- The system shall require the super admin to type a confirmation phrase before permanent deletion proceeds
- The system shall log all account actions with the acting admin's ID and timestamp

Merchant Communication

- The system shall allow the super admin to send a direct email message to any individual merchant
- The system shall log all communications sent to merchants from the admin panel

Non-Functional Requirements

- The merchant list shall load within 3 seconds even with 10,000+ merchant records
- Pagination shall display a maximum of 50 merchants per page to maintain performance
- Merchant account suspension shall take effect within 5 seconds of the super admin confirming the action
- All merchant management actions shall be written to the audit log immediately and shall be immutable
- Permanent merchant deletion shall be fully processed within 30 days in compliance with data privacy standards

Constraints & Edge Cases

- A super admin shall not be able to view or access a merchant's payment gateway credentials – they are encrypted and inaccessible even at admin level
- Permanently deleting a merchant cascades to remove shop data, products, and customization – however order records shall be retained for 1 year for financial compliance
- If a suspended merchant has active pending orders, those orders remain visible on customer tracking – but the merchant cannot log in to fulfill them
- A super admin cannot modify a merchant's product listings, prices, or order statuses – admin access is supervisory, not operational

A3

A3 — Platform Oversight & Analytics

Purpose:

This section defines the platform-wide data visibility and monitoring tools available to the super admin. These tools give the BazarHQ team real-time and historical understanding of platform performance and system health.

Actors Involved:

- Primary: Super Admin
- Secondary: BazarHQ Analytics Engine, System Health Monitor

Functional Requirements

Platform Overview Dashboard

- The system shall display a high-level platform overview dashboard as the super admin's home screen after login
- The dashboard shall display in real time: total registered merchants, total active shops, total orders today, total orders this month, and total platform revenue this month
- The dashboard shall display a graph showing new merchant registrations over the past 30 days
- The dashboard shall display a graph showing total platform orders over the past 30 days
- The dashboard shall highlight any unusual spikes or drops in key metrics compared to the previous period

Merchant Analytics

- The system shall display a ranked list of the top 10 highest-revenue merchants for the current month
- The system shall display the number of new merchants registered today, this week, and this month
- The system shall display the number of merchants who have published their shop versus those who have not completed onboarding
- The system shall display the churn rate — merchants who deleted their accounts within the past 30 days

Order & Revenue Analytics

- The system shall display total platform order count broken down by status across all merchants
- The system shall display total platform revenue broken down by payment method
- The system shall allow the super admin to filter all analytics by a custom date range
- The system shall allow the super admin to export a full platform analytics report as a CSV file

System Health Monitoring

- The system shall display the current operational status of all critical services: web server, database, storage, email gateway, and SMS gateway
- The system shall display server response time and error rate for the past 24 hours
- The system shall display a log of any system errors or failed operations from the past 7 days

- The system shall alert the super admin via email if any critical service goes down or error rate exceeds a defined threshold

Audit Log

- The system shall maintain a complete audit log of all super admin actions taken on the platform
- Each audit log entry shall record: action type, acting admin ID, target entity, timestamp, and IP address
- The system shall allow the super admin to search and filter the audit log by action type, admin ID, and date range
- The system shall retain audit logs for a minimum of 2 years
- The audit log shall be read-only — no entry can be modified or deleted by anyone including the super admin

Non-Functional Requirements

- The platform overview dashboard shall load within 3 seconds
- Platform-wide analytics data shall be aggregated and updated every 15 minutes
- The system health monitor shall check service status every 60 seconds and reflect changes within 2 minutes
- Audit log entries shall be written synchronously — an admin action shall not be considered complete unless its log entry is recorded
- Analytics exports for large date ranges shall be processed as background jobs and delivered via email

Constraints & Edge Cases

- Platform analytics are aggregated — the super admin sees totals and trends, not individual customer personal data
- If a merchant deletes their account, their historical contribution to revenue figures is retained in aggregates but their individual record is removed
- The system health monitor is informational only in Version 1 — it displays status but does not trigger automatic recovery
- Super admin analytics access is strictly read-only — no platform data can be modified from the analytics section

A4**A4 — Theme & Content Management****Purpose:**

This section defines how the super admin manages the storefront themes available to merchants and controls platform-level content such as announcements and policy pages.

Actors Involved:

- Primary: Super Admin
- Secondary: BazarHQ Theme Engine, All Merchant Storefronts

Functional Requirements**Theme Management**

- The system shall display a list of all available storefront themes with a preview image and the number of merchants using each
- The system shall allow the super admin to add a new theme by uploading its configuration and assets
- The system shall allow the super admin to update an existing theme's assets or configuration
- The system shall allow the super admin to preview any theme before making it available to merchants
- The system shall allow the super admin to set a theme as active (visible to merchants) or inactive (hidden from new selections)
- The system shall not allow the super admin to delete a theme currently in use by any merchants
- The system shall allow the super admin to designate one theme as the default applied to new shops during onboarding
- The system shall notify all merchants using a theme via email if that theme is scheduled for significant changes or deprecation

Platform Announcements

- The system shall allow the super admin to compose and send a broadcast announcement to all registered merchants
- The announcement shall be delivered to all merchants via email
- The announcement shall also appear as a notification banner inside the merchant dashboard until dismissed
- The system shall allow the super admin to schedule an announcement for a future date and time
- The system shall allow the super admin to view a history of all past announcements with delivery timestamp and recipient count
- The system shall allow the super admin to cancel a scheduled announcement before it is sent

Platform Content Management

- The system shall allow the super admin to update the BazarHQ platform landing page content
- The system shall allow the super admin to update the BazarHQ Terms of Service document
- The system shall allow the super admin to update the BazarHQ Privacy Policy document
- The system shall display the last updated date on all policy documents automatically upon saving

- The system shall allow the super admin to manage a Frequently Asked Questions (FAQ) section for merchants

Non-Functional Requirements

- Theme updates shall not cause downtime or visual breakage for merchants currently using that theme
- A theme preview shall load within 3 seconds in the admin panel
- Broadcast announcements shall be delivered to all merchant emails within 10 minutes regardless of merchant count
- Platform content updates shall reflect publicly within 60 seconds of the super admin saving changes
- All theme assets shall be stored and served via CDN for fast load times across all storefronts

Constraints & Edge Cases

- In Version 1, theme creation requires manual developer involvement — the admin panel manages which pre-built themes are available
- If a theme update introduces breaking changes, the super admin must notify affected merchants at least 7 days in advance
- Broadcast announcements cannot be recalled once sent — the super admin shall see a confirmation screen with recipient count before sending
- Changes to Terms of Service and Privacy Policy shall require a second super admin's approval before going live in Version 1

7. Glossary

Term	Definition
Merchant	A business owner who registers on BazarHQ, creates a shop, and manages products and orders
Customer	An end user who visits a merchant's storefront, browses products, and places orders
Super Admin	A BazarHQ team member with full administrative access to the entire platform
Shop Staff	A merchant's employee with limited dashboard access — deferred to Version 2
Multi-Tenancy	Architecture where multiple merchants share one platform instance but are fully data-isolated
Subdomain	A unique URL prefix assigned to each shop (e.g. myshop.bazarhq.com)
RBAC	Role-Based Access Control — system access is determined by the user's assigned role
JWT	JSON Web Token — a standard method for representing authentication claims securely
OAuth	Open Authorization — a protocol allowing users to log in using existing accounts (e.g. Google)
2FA	Two-Factor Authentication — an additional verification step beyond username and password
TOTP	Time-Based One-Time Password — a 2FA method generating short-lived codes via an authenticator app
bcrypt	A password hashing function used to securely store passwords
COD	Cash on Delivery — a payment method where the customer pays in cash upon receiving the order
SSLCommerz	Bangladesh's most widely used payment gateway aggregator
bKash	Bangladesh's most widely used mobile financial service
Nagad	A Bangladesh Postal Department-backed mobile financial service
CDN	Content Delivery Network — distributed server network for rapid delivery of static assets
API	Application Programming Interface — a defined communication interface between software systems
CSV	Comma-Separated Values — a plain text file format for importing and exporting tabular data
HTTPS	Hypertext Transfer Protocol Secure — encrypted communication protocol for web traffic
SaaS	Software as a Service — software hosted on the cloud and delivered via a web

Term	Definition
	browser
Storefront	The customer-facing public website of a merchant's shop on BazarHQ
Admin Panel	The private, authenticated dashboard used by a merchant or super admin
Onboarding	The guided setup process a new merchant completes after creating their shop
Variant	A specific version of a product defined by attributes such as size or color
Abandoned Cart	A shopping session where a customer added items to cart but did not complete checkout
Audit Log	An immutable, chronological record of all significant actions taken within the system
Churn Rate	The rate at which merchants leave the platform by deleting their accounts in a given period
RFC 6238	The internet standard defining the Time-Based One-Time Password (TOTP) algorithm used for 2FA

End of Document

This document is maintained by the BazarHQ Development Team. All requirements are subject to revision as the project evolves.